Retail Sales Analysis

Comprehensive SQL-driven insights on transaction data, customer behavior, and operational patterns

Project Scope

10K+

Transactions

Multi-year transaction dataset analyzed

1.5K+

Customers

Unique customer profiles tracked

100%

SQL-Driven

Comprehensive EDA and diagnostic analysis

Data Quality Foundation

Data Source

retail_db

Time Period

Multi-year coverage

Integrity

Missing values removed for clean analysis

Sales Performance Insights

Category Revenue

Significant dispersion across product categories

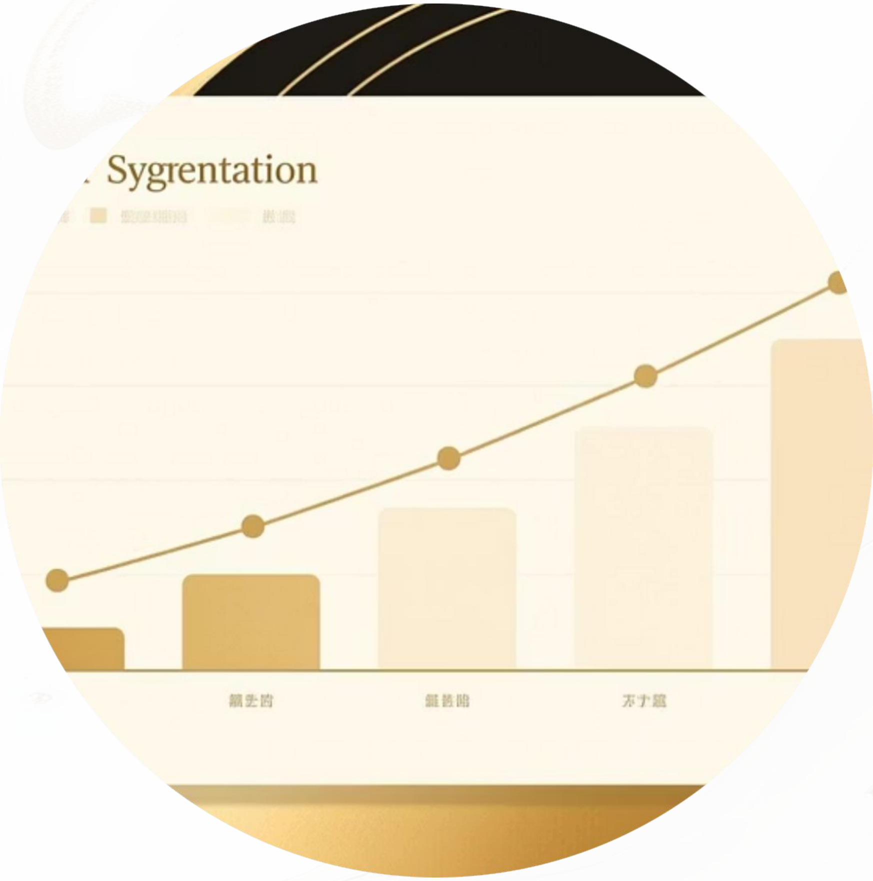
High-Value Transactions

Multiple transactions exceeding $1,000 identified

Performance Patterns

Monthly rankings reveal seasonal trends

Customer Behavior Patterns



Top Contributors

Top 5 customers drive major revenue share

Demographics

Clear patterns emerge by category preference

Purchase Behavior

Time-of-day impacts order frequency

Temporal Analysis

1 Monthly Ranking

Window functions reveal performance by month

2 Daily Patterns

Time-of-day analysis shows peak hours

3 Seasonal Trends

Multi-year view identifies cycles

SQL Techniques Applied

1

Aggregations

SUM, COUNT, AVG for core metrics

2

Window Functions

RANK, PARTITION BY for rankings

3

Date Functions

EXTRACT, TO_CHAR for temporal analysis

4

Data Cleaning

Validation queries ensure integrity

Business Recommendations

Loyalty Program

Launch program targeting top customers

Inventory Optimization

Align stock with time-of-day and monthly patterns

Targeted Marketing

Leverage demographic insights for campaigns

Limitations & Considerations

External Factors

- Holidays not captured
- Weather impact unknown
- Economic conditions excluded

Customer Insights

- No lifetime value data
- Customer journey gaps
- Limited behavioral depth

Future Directions

Time-Series Forecasting

Predict future sales trends

Customer Clustering

Advanced segmentation models

Real-Time Analytics

Live dashboards and alerts

Prepared By: Data Analytics Team